

Modeling Anthropogenic CO2 Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

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Abstract

The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO₂) concentrations. This report investigates whether the long-term increase in atmospheric CO₂ is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO₂ data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO₂ concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

1. Introduction

See code [1](#) in Appendix for the implementation.

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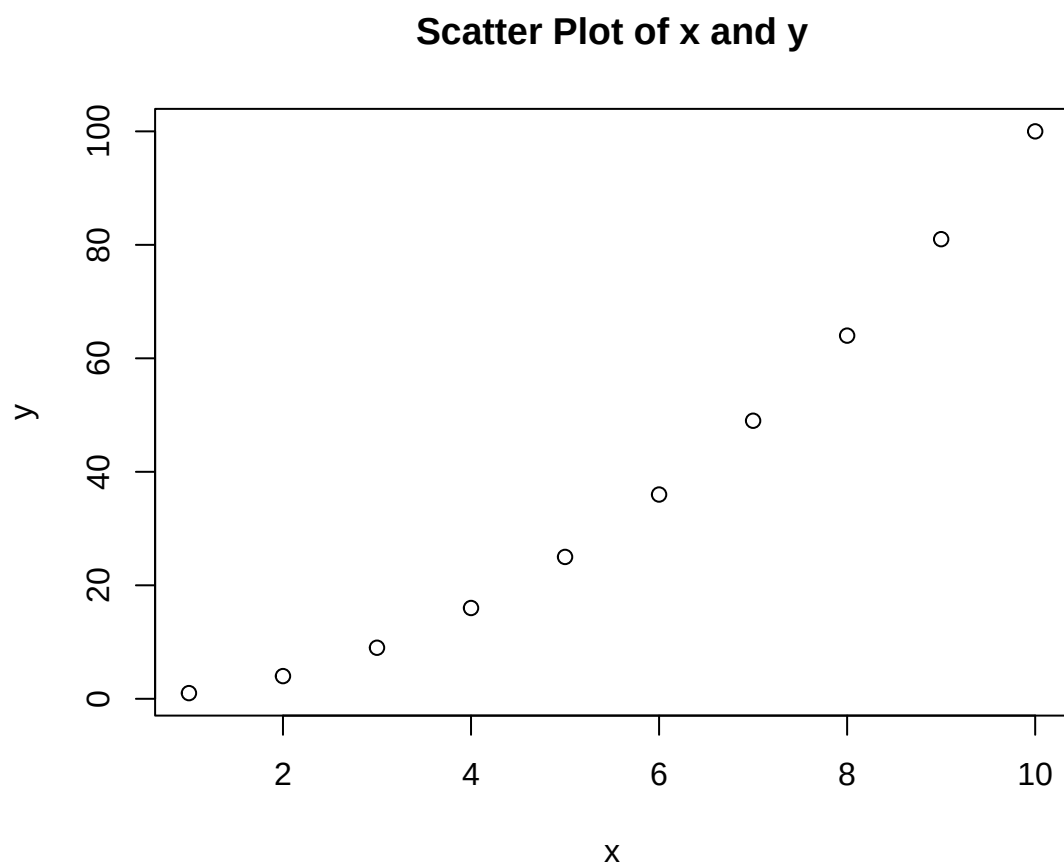


Figure 1: Scatter plot of the binary outcomes and the logistic regression curve

1.1. Scientific Context

1.2. Objectives

2. Data Description and Preprocessing

2.1. Dataset Overview

2.2. Data Cleaning

2.3. Aggregation

2.4. Exploratory Visualization

3. Bayesian Methodology and Models

3.1. Model 1: Linear Trend

3.2. Model 2: Quadratic Trend

3.3. Inference Strategy

4. Results and Inference

4.1. Parameter Estimation

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5. Predictive Forecasting

5.1. Ten-Year Projections

5.2. Uncertainty Quantification

6. Conclusion

6.1. Summary of Findings

6.2. Limitations and Future Work

6.3. Appendix: Source code

```
1 # This is an example of a test file for EDA (Exploratory Data Analysis)
2
3
4 # Load necessary libraries
5 x <- 1:10
6 y <- x^2
7 # Create a scatter plot
8 plot(x, y, main = "Scatter Plot of x and y", xlab = "x", ylab = "y")
```

Listing 1: Plotting the data